

Daily Geometry Drill #5

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: the polar of an external point · two equal tangents · $\sin \frac{\theta}{2} = \frac{r}{OP}$ · the director circle.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Read OP and r straight off, then the tangent length is one step of Pythagoras.

- Write down the length of the tangent from $(0, 5)$ to $x^2 + y^2 = 9$.
- The point $(0, 5)$ sits outside $x^2 + y^2 = 9$ with $OP = 5$ and $r = 3$. Write down $\tan \frac{\theta}{2}$, where θ is the angle between the two tangents.
- Write down the equation of the chord of contact (polar) drawn from the point $(5, 3)$ to $x^2 + y^2 = 25$.
- The circle $x^2 + y^2 = 25$ has director circle $x^2 + y^2 = 2r^2$. Write down the radius of that director circle.
- Write down the length of the tangent from $(7, 0)$ to $x^2 + y^2 = 36$.
- Two tangents from the point P to a circle of unknown radius make an angle of 60° and $OP = 6$. Write down the length of each tangent.
- Write down the equation of the chord of contact drawn from $(5, 0)$ to $x^2 + y^2 = 4$ (the two tangency points join in a vertical line).
- A tangent from the point P to $x^2 + y^2 = 25$ has length 12. Write down OP .
- The two tangents from P to the circle $x^2 + y^2 = 25$ make an angle of 120° . Write down OP .
- The chord of contact from the point P to $x^2 + y^2 = 16$ is the line $x + y = 4$. Write down P .

Section B Manipulation Drills

(≤50 s each)

Feed external points and radii through $\sqrt{OP^2 - r^2}$ and the polar formula, choosing from the options.

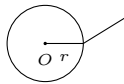
- The length of the tangent from $(3, 4)$ to $x^2 + y^2 = 9$ is?

A) $\sqrt{7}$

- B) 4
C) 5
D) 2
2. The point P has $OP = 6$ with respect to a circle of radius 3. The angle between the two tangents from P is?
A) 30°
B) 45°
C) 60°
D) 90°
3. Write down the equation of the chord of contact drawn from the point $(3, 5)$ to $x^2 + y^2 = 25$.
4. The director circle of $x^2 + y^2 = 4$ has radius?
A) 2
B) $2\sqrt{2}$
C) 4
D) $\sqrt{2}$
5. The length of the tangent from $(1, 2)$ to $x^2 + y^2 = 1$ is?
A) 1
B) $\sqrt{5}$
C) 2
D) $\sqrt{3}$
6. A tangent from P to a circle of radius 5 has length 12. How far is P from the centre?
A) 17
B) 7
C) 13
D) $\sqrt{119}$
7. The chord of contact from $P = (a, 0)$ to $x^2 + y^2 = 16$ is the line $x = 4$. Find a .
A) 2
B) 4
C) 8
D) 16
8. The two tangents from P to $x^2 + y^2 = 25$ meet at right angles (so P lies on the director circle). Write down the length of each tangent.

9. The tangent to $x^2 + y^2 = 8$ with slope 1 has positive y -intercept equal to?
- A) 4
B) $2\sqrt{2}$
C) 8
D) $4\sqrt{2}$
10. A tangent from P to a circle has length 6, and $OP = 10$. Write down the circle's radius.

Section C Substitution & Structure(≤ 75 s each)*Hard: equal tangents, director circle, and a hidden s – a trap. No direct plug.*

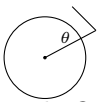
1. (after TMUA 2020 Paper 1 Q16 / BMO1 2013 Q4) From $P(8, 12)$, two tangents are drawn to $x^2 + y^2 = 36$ (centre O , $r = 6$). Let T_1T_2 be the chord of contact. What is the distance from O to the line T_1T_2 ?
- A) $\frac{9}{5}$
B) $\frac{18}{5}$
C) $\frac{36}{5}$
D) $\frac{9}{2}$
2. (adapted from TMUA 2018 Paper 1 Q7) The locus of points P where the tangent length to $x^2 + y^2 = 4$ equals the tangent length to $(x - 6)^2 + y^2 = 16$ is the radical axis. Which line is it?
- A) $x = 1$
B) $x = 2$
C) $x = 3$
D) $x = 4$
3.  (after MAT 2021 Q5) The circle $x^2 + y^2 - 6x - 8y + 24 = 0$ has centre $(3, 4)$, $r = 1$. The point $P(6, 8)$ is outside. What is the length of the tangent from P to the circle?
- A) 4
B) $2\sqrt{5}$
C) 5
D) $\sqrt{26}$
4. (adapted from BMO1 2015 Q3) For the 13-14-15 triangle ($s = 21$, $r = 4$), the two tangent lengths from the vertex opposite side 14 are $s - b$. With $b = 14$, what is that length?
- A) 6
B) 7
C) 8
D) 9

5. (after TMUA 2022 Paper 1 Q19) The circle $(x - 4)^2 + y^2 = 9$ has director circle $x^2 + y^2 = 2r^2$ for a centred circle, but this circle is translated. What is the locus of points from which the two tangents to $(x - 4)^2 + y^2 = 9$ are perpendicular?
- A) $(x - 4)^2 + y^2 = 18$
 B) $(x - 4)^2 + y^2 = 9$
 C) $x^2 + y^2 = 18$
 D) $x^2 + y^2 = 9$
6.  (adapted from TMUA 2020 Paper 1 Q16) The point $(8, 6)$ is at distance 10 from the origin. Tangents from $(8, 6)$ to $x^2 + y^2 = 1$ have length $\sqrt{99}$. What is the distance from the origin to the chord of contact?
- A) $\frac{1}{10}$
 B) $\frac{1}{\sqrt{101}}$
 C) $\frac{1}{\sqrt{99}}$
 D) $\frac{10}{\sqrt{101}}$
7. (after BMO1 2020 Q3) The triangle with sides 13, 14, 15 has incircle touching BC at D . If $BD = 6$, $DC = 9$, and $s = 21$, the two equal tangents from A are $s - a$. What is AE (tangents from A)?
- A) 6
 B) 7
 C) 8
 D) 9
8. (adapted from TMUA 2021 Paper 2 Q7) The chord of contact of $(2, 1)$ w.r.t. $x^2 + y^2 = 1$ is $2x + y = 1$. What is its distance from the origin, and is it inside, on, or outside the circle?
- A) $\frac{1}{\sqrt{5}}$ (inside, $d < r$)
 B) $\frac{1}{\sqrt{5}}$ (outside, $d > r$)
 C) $\frac{\sqrt{5}}{5}$ (inside)
 D) 1 (tangent)

Section D Challenge Ramp

(≤ 90 s each)

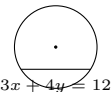
Hardest 5: angle $\theta/2$ via r/OP , chord AB via power of a point, and a hidden incircle.

1.  (after TMUA 2018 Paper 1 Q7 / BMO1 2023 Q4) Two tangents from P to $x^2 + y^2 = 9$ with $OP = 6$ enclose angle θ . What is θ ?
- A) 30°
 B) 60°

- C) 90°
 D) 120°

2. (adapted from TMUA 2020 Paper 1 Q16) From P to $x^2 + y^2 = 25$, tangent length is $5\sqrt{3}$. Find OP (use $OP^2 = r^2 + \text{tangent}^2$).

- A) 10
 B) $5\sqrt{10}$
 C) $10\sqrt{3}$
 D) 75

3.  (adapted from TMUA 2019 Paper 1 Q6) The chord of contact from $P(3, 3)$ to $x^2 + y^2 = 9$ is $x + y = 3$. What is its length? (Hint: distance from O to chord is $3/\sqrt{2}$, $r = 3$)

- A) $3\sqrt{2}$
 B) $\frac{3\sqrt{14}}{2}$
 C) $3\sqrt{6}$
 D) 6

4. (after BMO1 2015 Q3) Two tangents from P make 60° and $OP = 10$, $r = OP \sin(\theta/2) = 5$, tangent length = $OP \cos(\theta/2)$. What is the length?

- A) 5
 B) $5\sqrt{3}$
 C) 10
 D) $5/2$

5.  (adapted from MAT 2021 Q5) The polar of $(9, 12)$ w.r.t. $x^2 + y^2 = 81$ is $3x + 4y = 27$. What is its distance from the origin, and is P outside/inside/on?

- A) $27/5$ (outside, $d < r$? No, $d = 27/5 = 5.4 < r = 9$? Actually $r = 9$, $d = 5.4 < r$, so chord inside)
 B) $27/15$ (outside)
 C) $9/5$ (inside)
 D) $81/15$ (outside)

“A tangent is an archer’s draw: Pythagorean strength $OP^2 = r^2 + (\text{tangent})^2$ fired down a chord of contact that is just the same formula straightened.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths